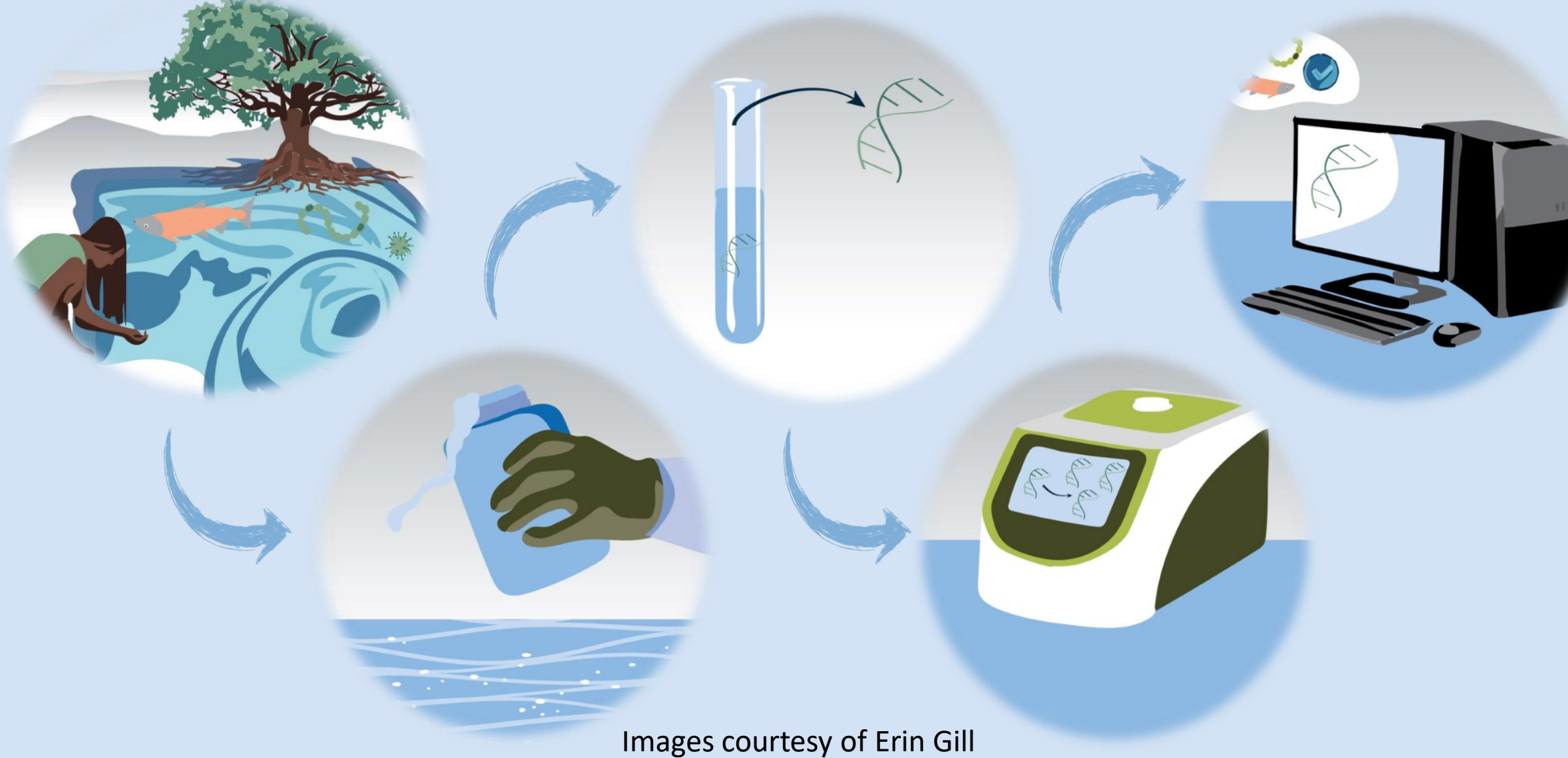


iMicroSeq: Integrated Resources Supporting Clinical and Environmental Sequence Data

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eDNA/eRNA Monitoring

Environmental DNA (eDNA) and Environmental RNA (eRNA) monitoring is a non-invasive monitoring technique that identifies organisms anywhere by analyzing trace genetic material shed into the environment.



Images courtesy of Erin Gill

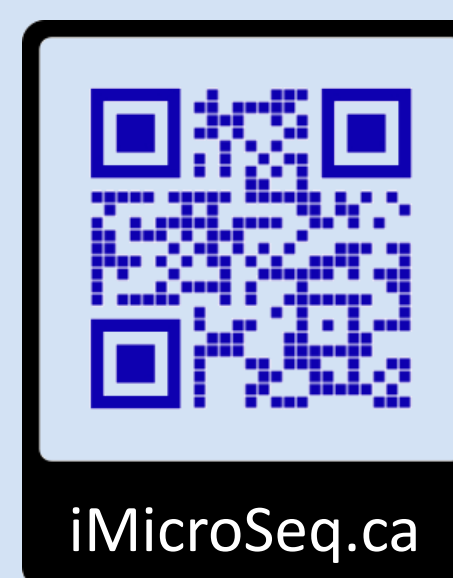
iMicroSeq

Inclusive, Integrative Microbial Sequencing

An inclusive framework enabling robust microbial/eDNA sequence data linkage, discovery, and analysis with relevant environmental data to protect health and environmental wellbeing across Canada

What iMicroSeq provides:

- Data discovery, uniting data in Canada into a standardized, searchable resource.
- More wholistic analyses of data from diverse sources, breaking down data silos.
- Integration with, and extending, data from other resources (e.g. GenBank, ENA), via the use of powerful data specifications and interchange formats.
- Supporting: Detection, source attribution, surveillance, risk assessment, and control strategies for microbes and antimicrobial resistance, bolstering sectors as diverse as public health, agriculture, aquaculture, and mining.



iMicroSeq.ca

ChùNet

Indigenous-led Local Knowledge and Data Sharing Network

An Indigenous co-led project that aims to empower community-led learning from water, guided by both genomics and First Nation Ways of Knowing.

Key Activities Include:

- Raising awareness of the benefits of eDNA monitoring through culturally appropriate media and outreach.
- Building First Nation partnerships focused on co-developing eDNA monitoring programs that address community priorities and local challenges.
- Training community members to independently conduct eDNA monitoring and environmental stewardship.
- Returning monitoring results to communities using preferred and locally appropriate communication methods.
- Developing data sharing agreements that reflect OCAP®



Funding & Affiliation



Acknowledgement

We thank all of the over 80 iMicroSeq participants for their valuable input.

Land Acknowledgement

We acknowledge this work unfolds across the unceded traditional territories of First Nations, Métis and Inuit Peoples, from coast to coast to coast. We honour their enduring relationships with land, water, and all life. Our work will be informed by active consultations with Indigenous communities. We further accept the programs' responsibilities to uphold Indigenous rights, move forward with Reconciliation, form respectful partnerships, and ensure our work supports the well-being of present and future generations.



Scan For Copy

The iMicroSeq Data Portal

Early Access: This is a preview version of the iMicroSeq Data Portal, currently under active development and open consultation. For updates, reporting problems, or to contribute data, contact info@imicroseq-dataportal.ca.



The **iMicroSeq Data Portal** is an inclusive, Canadian database of microbial sequences, and associated harmonized contextual metadata from clinical and environmental sources, including wastewater, to monitor, understand, and respond to threats, affecting public health, agriculture, aquaculture, and the environment.

- Powered by OICR's Overture Framework
- Metadata harmonization with Ontologies
- Visualizations and modelling tools
- Data type stored: qPCR, metagenomic, water, wastewater, antimicrobial resistance, chemistry, much more.



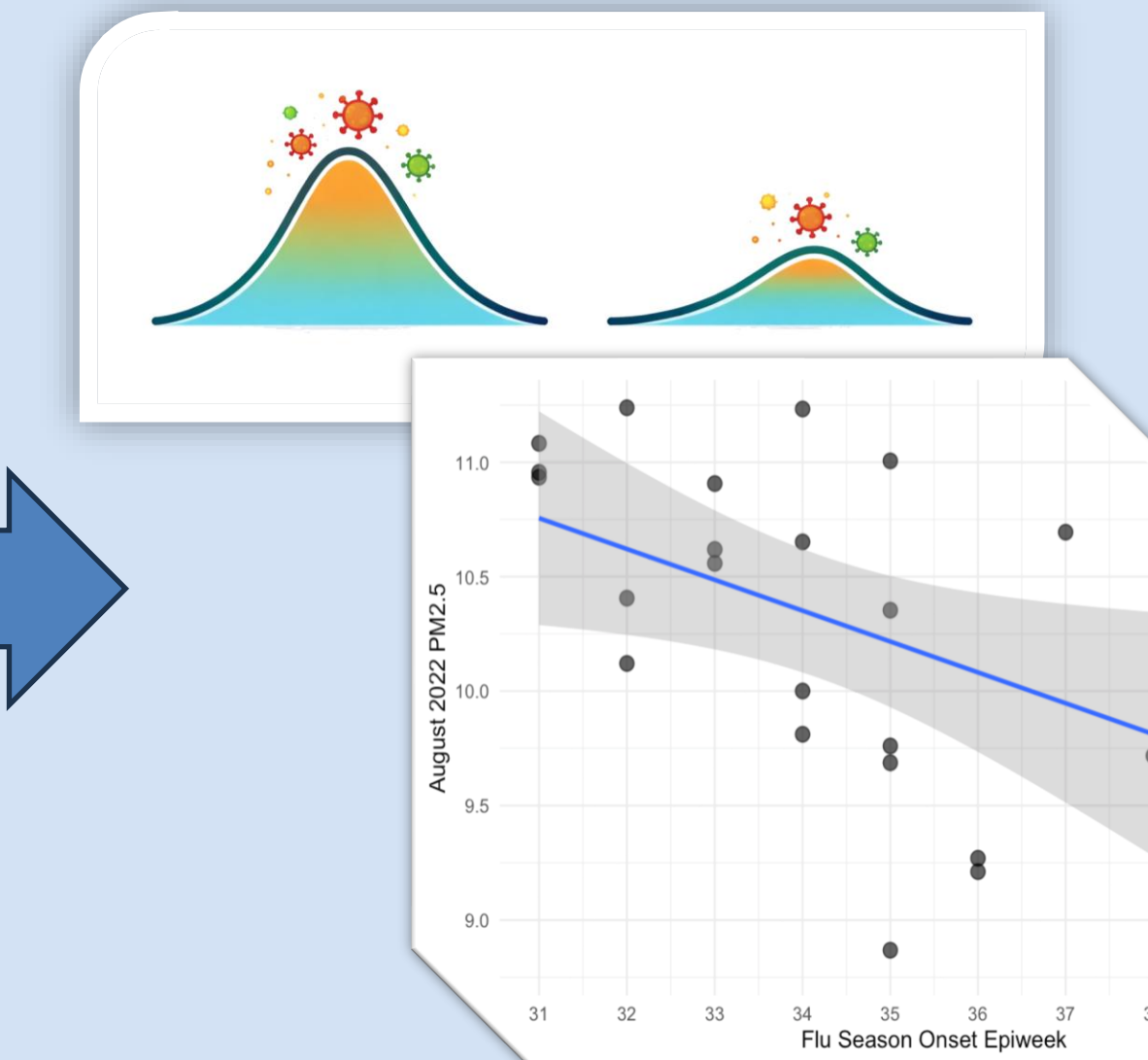
Impacts & Applications: Integrative Analysis

Multi-source data integration



- Environmental/clinical sample data,
- Environmental data
- Demographic and socioeconomic data
- Much more

Detection and Pattern discovery



- Spatio-temporal patterns of pathogens and other microbes

Actionable insights

- Early warning and outbreak detection
- Targeted public health interventions
- Agricultural and environmental health interventions
- Policy and infrastructure planning
- Communities informed and empowered

Images courtesy of One Yukon Coalition, Nima Tehrani

Governance & Equity

iMicroSeq commits to:

- FAIR Principles
- Data security & privacy
- Recognition of data submitters
- Open standards & metadata specifications
- Government-academic collaboration
- Data governance transparency
- Respecting Indigenous data sovereignty
- Canadian sovereignty

Interested in learning more, stay updated, or to contribute data?

Email info@imicroseq.ca

